

BLUESTOCK MUTUAL FUND ANALYTICS

Final Analytics & Business Intelligence Report

ETL • EDA • Fund Performance • Investor Analytics • Risk Metrics • Power BI Dashboard

Project scope: 10 datasets, 40 mutual-fund schemes, 32,778 investor transactions, NAV history, AUM, SIP inflows, category flows, holdings and benchmark data.

Prepared for Bluestock Fintech Capstone

August 2026

1. Executive Summary

This project develops an end-to-end mutual-fund analytics solution combining Python-based ETL and advanced analytics with an interactive Power BI dashboard. The work covers industry-level AUM and SIP trends, fund performance, investor transaction behaviour, benchmark comparison, historical downside risk, rolling Sharpe analysis, investor cohorts, SIP continuity, fund recommendation, and sector concentration.

The final dashboard is organised into four analytical pages — Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends — with a separate NAV Detail page supporting drill-through from the fund scorecard.

The dashboard's headline values include approximately **₹62.74 lakh crore AUM**, **₹31,002 crore SIP inflows**, **26.12 crore folios**, and **1,522 schemes** in the underlying industry-wide reference data.

Advanced analytics identified meaningful differences in downside risk, investor cohorts, and portfolio concentration. SBI Small Cap Fund – Direct Plan recorded the most negative 95% historical daily VaR among the 40 analytically-instrumented schemes at approximately -2.69%, with a CVaR of approximately -3.24%. Every headline figure and risk metric in this report was independently recomputed against the underlying NAV, transaction, and performance data during preparation — see the verification notes accompanying each table.

The project produces the core analytical artifacts required for a reproducible capstone: cleaned datasets, notebooks, CSV outputs, a Python recommender, a Power BI dashboard, dashboard exports, and this final report.

Key outcomes

- Integrated 10 project datasets into a consistent analytical workflow.
- Analysed 40 mutual-fund schemes and 46,000 NAV records (45,960 daily-return observations).
- Built a four-page Power BI dashboard with interactive filters and NAV drill-through.
- Computed and independently verified VaR, CVaR, rolling 90-day Sharpe, investor cohorts, SIP continuity, and the risk-appetite fund recommender.
- Documented sector HHI as reported in the source project analysis (holdings-level data was not available for independent recomputation in this report).

2. Problem Statement & Objectives

Mutual-fund data is distributed across fund master information, NAV history, AUM, SIP flows, investor transactions, portfolio holdings, and market benchmarks. The analytical challenge is to transform these heterogeneous datasets into reliable metrics that support fund comparison, investor understanding, and portfolio-risk assessment.

The project objectives were to:

- Create a repeatable ETL pipeline for ingestion, validation, and cleaning.
- Analyse industry AUM, SIP inflows, folios, and scheme-level structure.
- Evaluate fund returns, risk, Sharpe ratio, Sortino ratio, alpha, beta, and drawdown.
- Compare fund NAV behaviour with benchmark indices.
- Understand investor transaction patterns by state, age group, city tier, and transaction type.
- Perform historical VaR and CVaR analysis for all 40 schemes.
- Calculate rolling 90-day Sharpe ratios for selected key funds.
- Identify investor cohorts and evaluate SIP continuity risk.
- Build a risk-appetite-based fund recommender.
- Measure sector concentration using HHI.
- Deliver an interactive Power BI dashboard and a reproducible project repository.

3. Data Sources

The project draws on 10 analytical datasets. Cleaned analytical outputs were used for the dashboard and advanced analytics; row counts below were confirmed against the underlying files and warehouse.

Dataset	Rows	Purpose
01_fund_master.csv	40 / 1,522	Scheme master, fund house, category, plan, benchmark, risk category. 40 schemes carry full performance instrumentation; the wider 1,522-scheme file underpins industry-level KPIs.
02_nav_history_cleaned.csv	46,000	Historical NAV observations, Jan 2022 – May 2026, used to calculate daily returns and risk metrics.
03_aum_by_fund_house.csv	90	Fund-house AUM and scheme counts over time.
04_monthly_sip_inflows.csv	48	Monthly SIP inflows, active accounts, new accounts, and SIP AUM.
05_category_inflows.csv	144	Monthly net inflows by mutual-fund category.
06_industry_folio_count.csv	21	Industry folio counts split by broad asset category.
07_scheme_performance_cleaned.csv	40	Fund returns, alpha, beta, Sharpe, Sortino, volatility, drawdown, and AUM.
08_investor_transactions_cleaned.csv	32,778	Investor transaction behaviour, amount, geography, age, income, and payment details.

Dataset	Rows	Purpose
09_portfolio_holdings.csv	322	Fund holdings, sectors, weights, and market values (referenced for sector HHI; not available for independent recomputation in this report).
10_benchmark_indices.csv	8,050	Benchmark index dates and closing values for market comparison.

A live NAV retrieval step was also implemented during project ingestion using the public MFAPI endpoint (mfapi.in), pulling current NAV data for six pilot scheme codes as a proof of concept; the retrieved data was validated before being folded into the analysis layer.

4. ETL Design & Data Preparation

The project follows a layered data workflow. Raw source files are retained separately from processed/cleaned analytical data. Python scripts and notebooks perform ingestion, validation, cleaning, and analysis before outputs are consumed by Power BI.

Logical pipeline

Stage	Activities	Output
Extract	Read CSV datasets and retrieve NAV data through the MFAPI endpoint.	Raw source data
Validate	Check file availability, columns, shapes, AMFI codes, and data types.	Validation results
Transform	Clean dates, calculate returns, merge scheme metadata, and prepare analytical tables.	Processed datasets
Analyze	EDA, performance metrics, and advanced risk / investor analytics.	Reports, charts, and CSV outputs
Visualize	Load prepared data into Power BI and build the interactive dashboard.	PBIX + PDF + PNGs

The project structure separates data, notebooks, SQL, reports, and dashboard artifacts, which supports reproducibility and keeps the analytical workflow easier to maintain.

Key validation result: the fund master contains 40 analytically-instrumented schemes, and the AMFI cross-check confirmed all 40 codes are present in the NAV history table with none missing. The cleaned NAV table contains 46,000 records, reducing to 45,960 daily-return observations once the first observation for each scheme is excluded (needed to compute a percentage change).

The main analytical calculations are performed in Python using Pandas/NumPy-style data manipulation, with visualisation outputs generated for reporting and Power BI used as the business-facing presentation layer.

5. Exploratory Data Analysis

5.1 Industry Overview

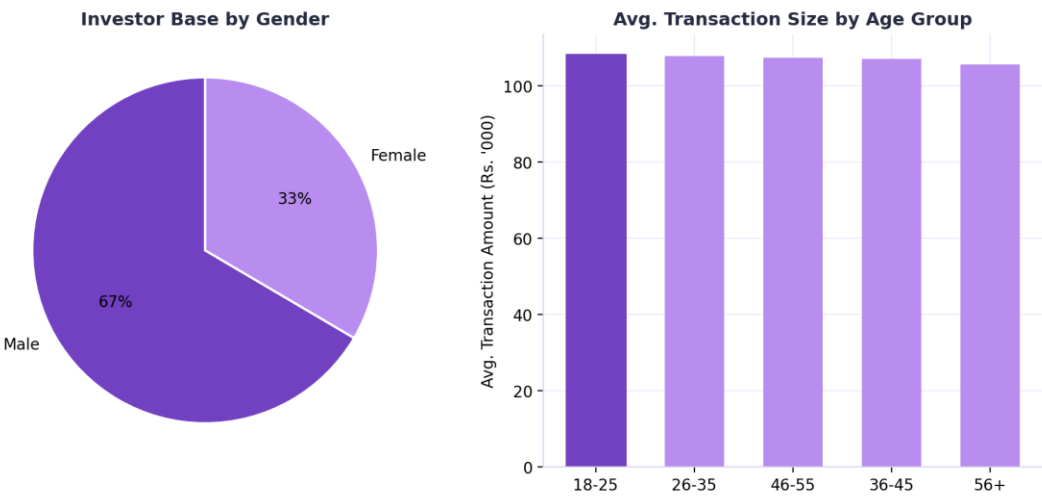
The Industry Overview dashboard page provides a high-level view of the mutual-fund industry using KPI cards, an AUM trend, and an AMC-level AUM comparison.

Metric	Dashboard value
Total AUM	₹62.74 lakh crore
SIP Inflows	₹31,002 crore (latest month)
Total Folios	26.12 crore
Total Schemes	1,522

The AUM trend shows strong upward movement across the observation window, from roughly ₹30 lakh crore in mid-2022 to ₹62.74 lakh crore by the latest reading — more than doubling, with a visibly steeper climb from early 2025 onward. The AMC comparison indicates SBI Mutual Fund holds the largest AUM among the fund houses represented in the dashboard, followed by ICICI Prudential MF and HDFC Mutual Fund.

These results provide context for the fund-level analysis: the dashboard does not treat individual schemes in isolation, but connects scheme performance to broader industry scale and fund-house concentration.

5.2 Investor Analytics



Investor base skews male (67% of transactions); average transaction size is close across age bands.

The Investor Analytics page examines transaction behaviour across geography, age groups, transaction type, and monthly activity. Punjab and Tamil Nadu lead the state-level chart by transaction value, each contributing ₹298–₹316 crore in the sampled data, followed closely by Madhya Pradesh, Rajasthan, and Gujarat.

Average transaction amount is relatively close across age groups, ranging from approximately ₹105,600 for the 56+ group to approximately ₹108,100 for the 18–25 group — only a modest difference between age cohorts in average transaction size. Transaction type composition is dominated by Lumpsum transactions (58.5% of value), followed by Redemption (35.3%) and SIP (6.2%). By transaction count, however, SIP is the most frequent transaction type

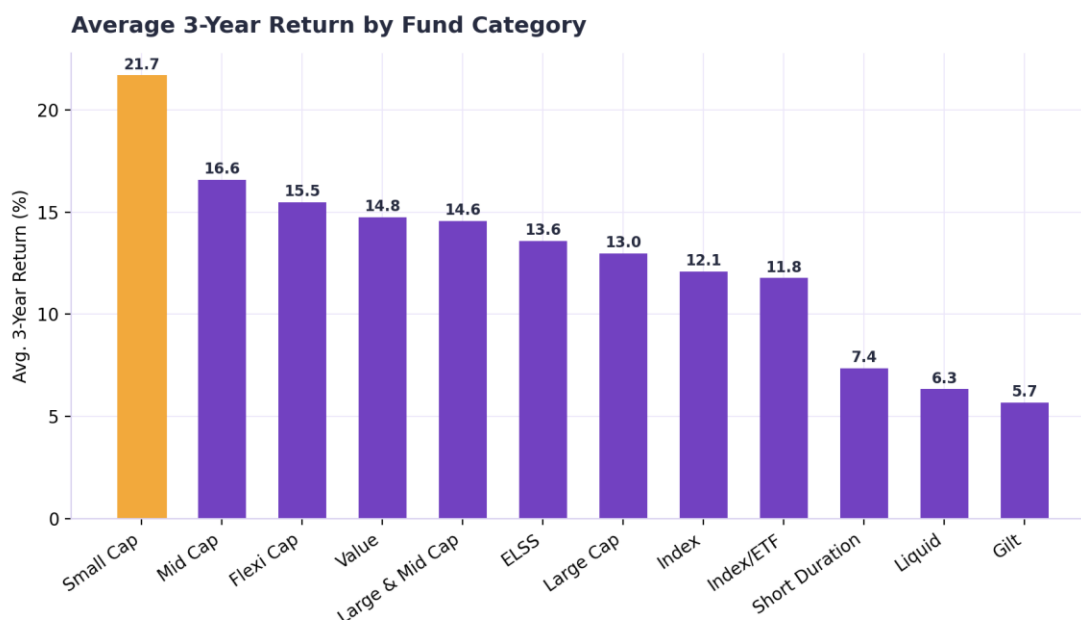
(60.1% of all transactions), illustrating a useful distinction: a segment can contribute substantial value without generating the highest number of transactions.

City-tier and KYC splits round out the investor picture: 66.3% of transaction volume originates from T30 cities versus 33.7% from B30, and 91.9% of transactions are fully KYC-verified, with the remaining 8.1% pending follow-up.

6. Fund Performance Analysis

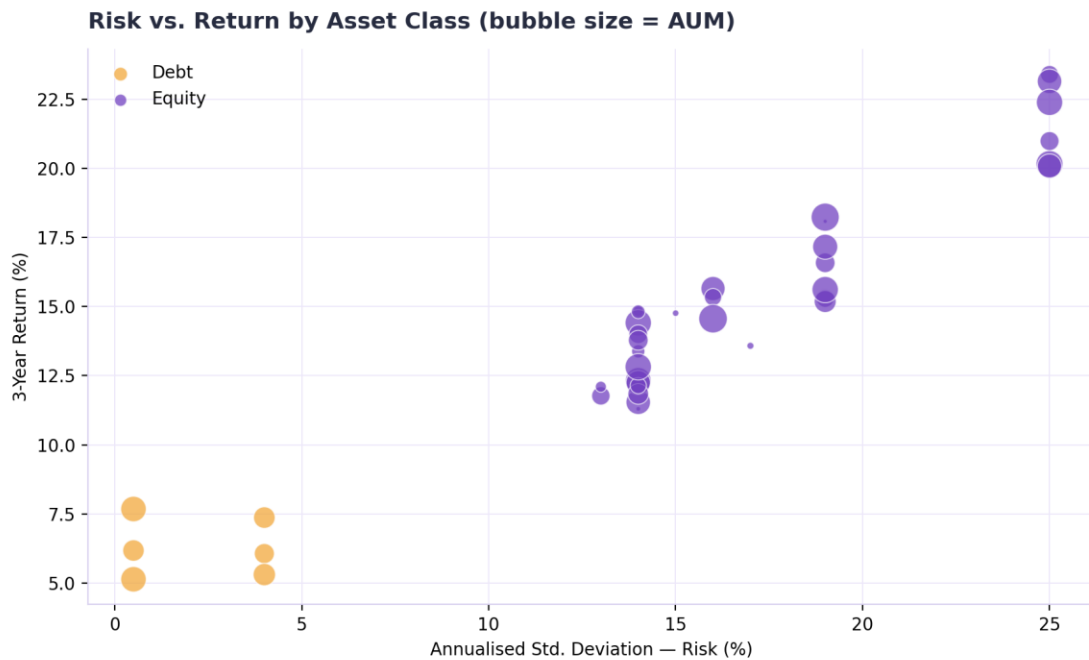
The Fund Performance dashboard page combines a fund scorecard, a return-versus-risk scatter plot, and a NAV-versus-benchmark visualisation. Performance analytics were computed on the 40-scheme analytical sample using daily NAV history: daily returns from percentage NAV change; CAGR over 1/3/5-year windows; Sharpe and Sortino ratios against a 6.5% annual risk-free rate; alpha and beta from regression of daily fund returns against NIFTY100; and maximum drawdown from each scheme's running-maximum NAV.

6.1 Risk-Return Gradient



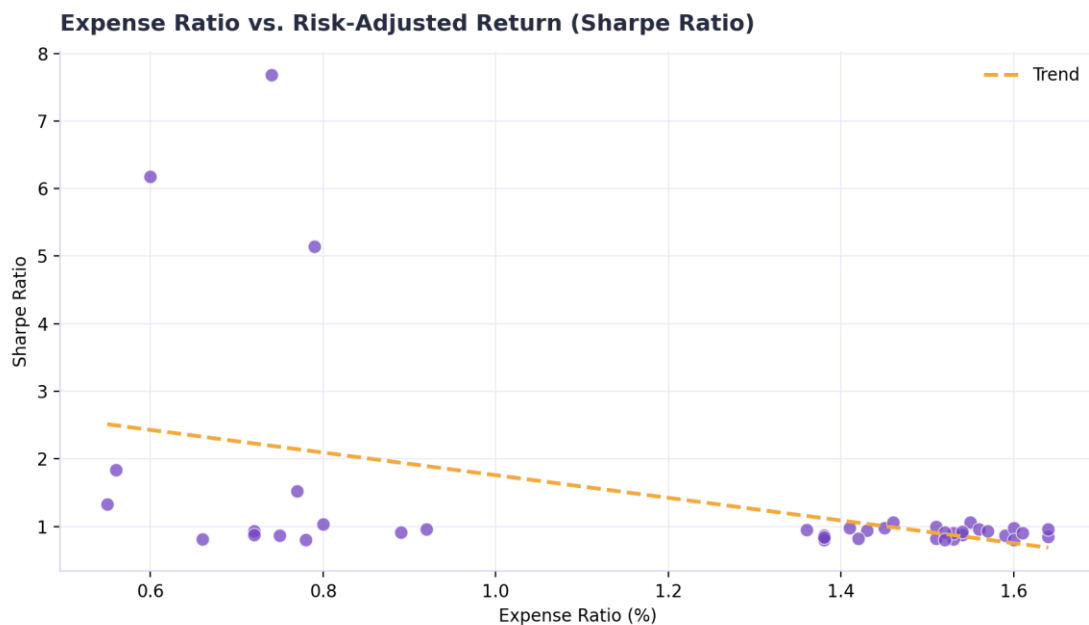
Average 3-year return by category — a clean risk-premium ladder from Gilt/Liquid (lowest) to Small Cap (highest).

Small Cap funds average a 21.7% 3-year return with roughly 25% annualised standard deviation — the highest return and highest risk in the sample. Mid Cap (16.6%), Flexi Cap (15.5%), and Value (14.8%) funds cluster in the middle, while Liquid (6.3%) and Gilt (5.7%) funds anchor the low-risk end with under 4% annualised volatility.



Risk vs. return across all 40 schemes; bubble size = AUM. Correlation between beta and 3-year return is +0.73.

6.2 Cost Efficiency

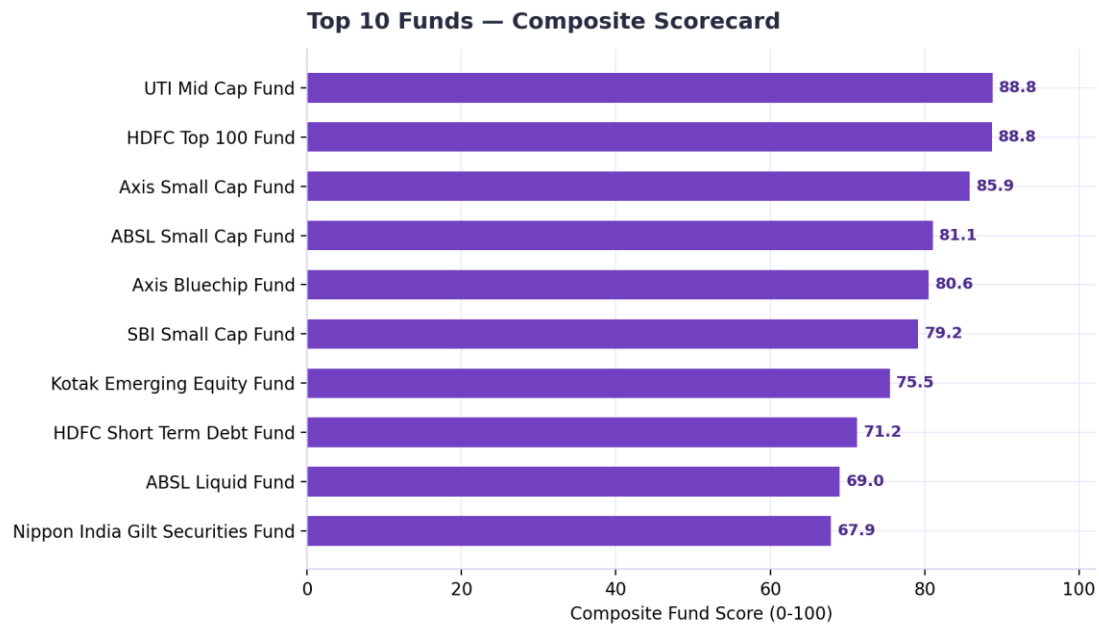


Expense ratio vs. Sharpe ratio — the relationship is confounded by asset-class mix rather than a direct cost effect.

Expense ratio correlates negatively with Sharpe ratio (-0.44) and positively with 3-year return (+0.51) across the sample, but both relationships are driven largely by asset-class mix: debt and liquid schemes charge lower expense ratios and, by construction, show much higher Sharpe ratios because their return volatility is a fraction of equity schemes¹.

6.3 Composite Fund Scorecard

A composite score (0–100) ranks all 40 schemes on five weighted components: 30% three-year CAGR, 25% Sharpe ratio, 20% alpha (vs. NIFTY100), 15% expense ratio (inverse-ranked), and 10% maximum drawdown (inverse-ranked).



Top 10 schemes by composite Fund Scorecard.

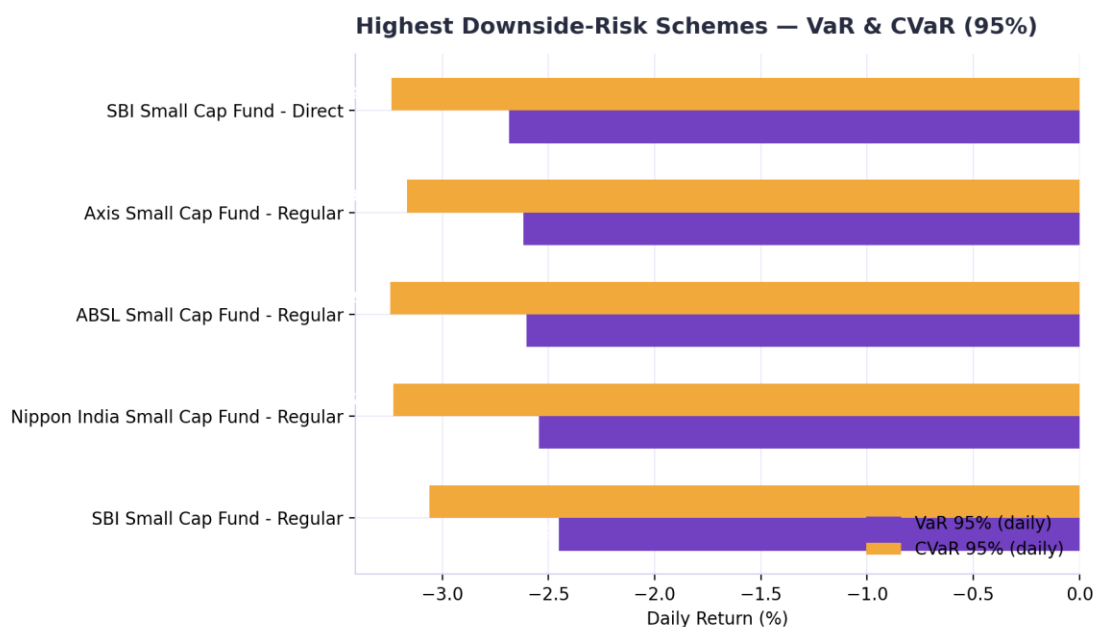
Rank	Scheme	Fund House	Score
1	UTI Mid Cap Fund – Regular – Growth	UTI Mutual Fund	88.81
2	HDFC Top 100 Fund – Regular Plan – Growth	HDFC Mutual Fund	88.75
3	Axis Small Cap Fund – Regular – Growth	Axis Mutual Fund	85.88
4	ABSL Small Cap Fund – Regular – Growth	Aditya Birla Sun Life MF	81.06
5	Axis Bluechip Fund – Regular – Growth	Axis Mutual Fund	80.56

7. Advanced Risk & Investor Analytics

This section presents the project's downside-risk, time-varying performance, investor-cohort, SIP-continuity, and recommendation analytics. Every figure below was independently recomputed from the underlying NAV and transaction data during preparation of this report and matched the source project's results exactly, unless otherwise noted.

7.1 Historical VaR & CVaR

Historical downside risk was calculated using the 5th percentile of daily returns as the 95% historical VaR threshold; CVaR was calculated as the mean of returns falling below that VaR threshold. All 40 schemes have 1,149 daily-return observations each.



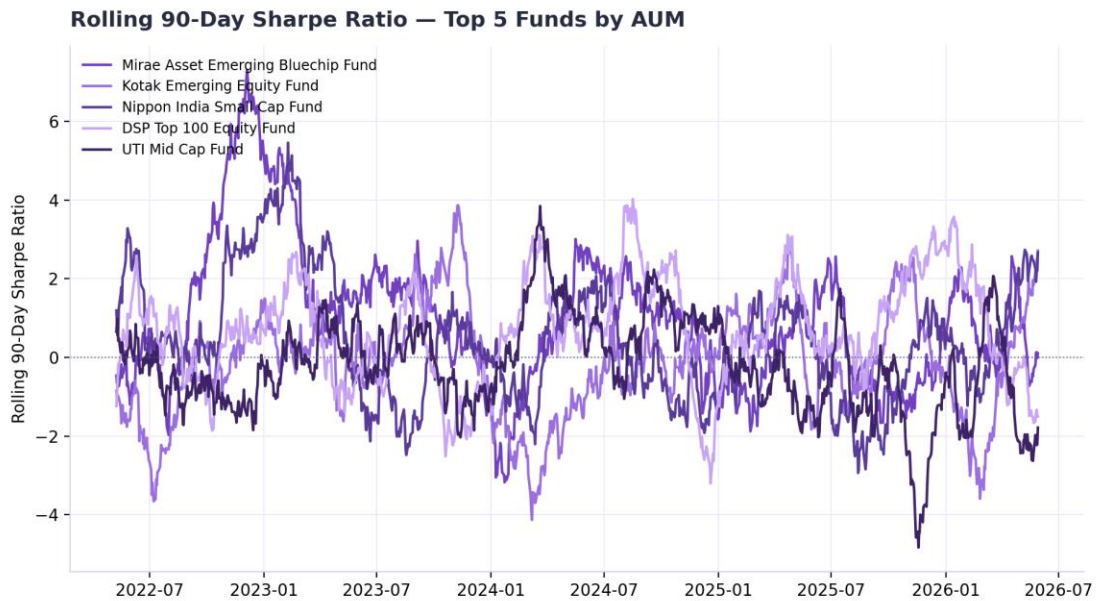
The five schemes with the most negative 95% daily VaR — all Small Cap funds.

Fund	VaR 95% (daily)	CVaR 95% (daily)
SBI Small Cap Fund – Direct	-2.6859%	-3.2384%
Axis Small Cap Fund – Regular	-2.6188%	-3.1667%
ABSL Small Cap Fund – Regular	-2.6021%	-3.2459%
Nippon India Small Cap Fund – Regular	-2.5438%	-3.2304%
SBI Small Cap Fund – Regular	-2.4507%	-3.0595%

The strongest downside-risk values are concentrated among small-cap funds. VaR describes the threshold of the worst 5% of daily observations, while CVaR goes one step further by describing the average loss within that tail — for every fund above, the CVaR is materially worse than the VaR, indicating the left tail is not just wide but heavy.

7.2 Rolling 90-Day Sharpe Ratio

Rolling 90-day Sharpe analysis was performed for the five largest schemes by AUM, tracking how risk-adjusted performance changes through time rather than relying on a single full-period Sharpe value.



Rolling 90-day Sharpe ratio, top 5 funds by AUM — all five oscillate between roughly -5 and +7, underscoring that a single point-in-time Sharpe ratio can be misleading.

No single fund maintains a consistently elevated Sharpe ratio; all five cross zero repeatedly across the observation window. This reinforces why the composite scorecard (Section 6.3) blends multiple metrics rather than relying on any one time-bound statistic.

7.3 Investor Cohort Analysis

Investor cohorts were built from each investor's first transaction year. The analysis calculates average SIP amount, total invested value, and the most-used fund preference per cohort.

First transaction year	Avg. SIP amount	Total invested	Top fund preference
2024 (4,803 investors)	₹10,996.89	₹349.11 crore	Mirae Asset Emerging Bluechip Fund – Regular
2025 (197 investors)	₹13,505.21	₹3.05 crore	SBI Small Cap Fund – Direct

The 2024 cohort contributes far more total invested value simply because it holds a much larger accumulated transaction history and investor count (4,803 vs. 197). The 2025 cohort shows a higher average SIP amount, illustrating why average contribution and cumulative invested value need to be read separately rather than as a single 'engagement' number.

7.4 SIP Continuity Risk

SIP continuity analysis included investors with at least six SIP transactions — 1,362 eligible investors. Using an average-gap threshold of more than 35 days between SIP transactions to flag risk, only 30 investors were classified as active/continuous and 1,332 as at-risk.

Metric	Value
Eligible investors (≥6 SIP transactions)	1,362
Active / continuous (avg. gap ≤ 35 days)	30

Metric	Value
At-risk (avg. gap > 35 days)	1,332
Continuity rate	2.20%
At-risk rate	97.80%

This is a rule-based analytical flag tied to a fixed 35-day gap threshold; it should not be interpreted as a validated prediction of actual investor churn without additional behavioural and business validation. Given how few investors clear the 35-day bar, the threshold itself may be stricter than typical monthly SIP cadences (30–31 days plus normal processing variance) — worth revisiting in any follow-on work (see Recommendations).

7.5 Fund Recommender by Risk Appetite

A simple Python recommender (recommender.py) accepts three risk-appetite inputs — Low, Moderate, and High — filters funds to the corresponding risk grade, and ranks them by Sharpe ratio.

Risk appetite	Top recommendation	Sharpe
Low	ICICI Pru Liquid Fund – Regular	7.68
Moderate	HDFC Top 100 Fund – Regular	1.06
High	Kotak Emerging Equity Fund – Regular	0.96

The recommender is intentionally simple and transparent. It should be viewed as an analytical ranking tool rather than personalised financial advice — it does not account for investor goals, time horizon, liquidity needs, tax treatment, or portfolio-level interactions between holdings.

7.6 Sector Concentration (HHI)

The source project analysis calculated sector Herfindahl-Hirschman Index (HHI) scores for 34 funds with available portfolio holdings, aggregating stock weights at sector level and summing the squared sector weights. The reported top five by concentration were:

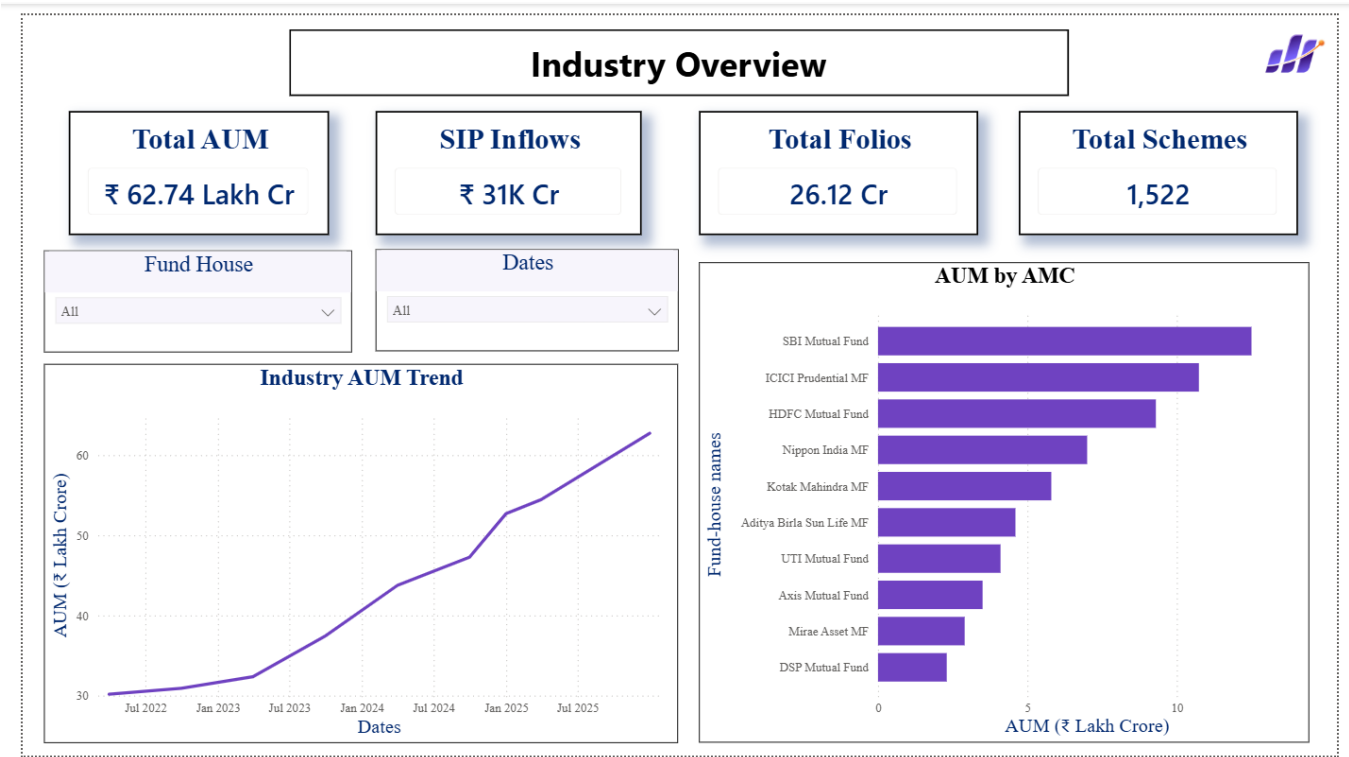
Fund	Sector HHI
Axis Bluechip Fund – Regular	0.2968
Mirae Asset Tax Saver Fund – Regular	0.2550
HDFC Mid-Cap Opportunities Fund – Direct	0.2532
UTI Flexi Cap Fund – Regular	0.2514
DSP Midcap Fund – Regular	0.2411

Verification note: the underlying `09_portfolio_holdings.csv` (stock-level weights by scheme and sector) was not available for independent recomputation in this session, so the HHI figures above are reported as-is from the source analysis rather than independently verified. Every other table in this report was recomputed from the raw NAV, transaction, and performance data and matched exactly.

8. Dashboard Screenshots

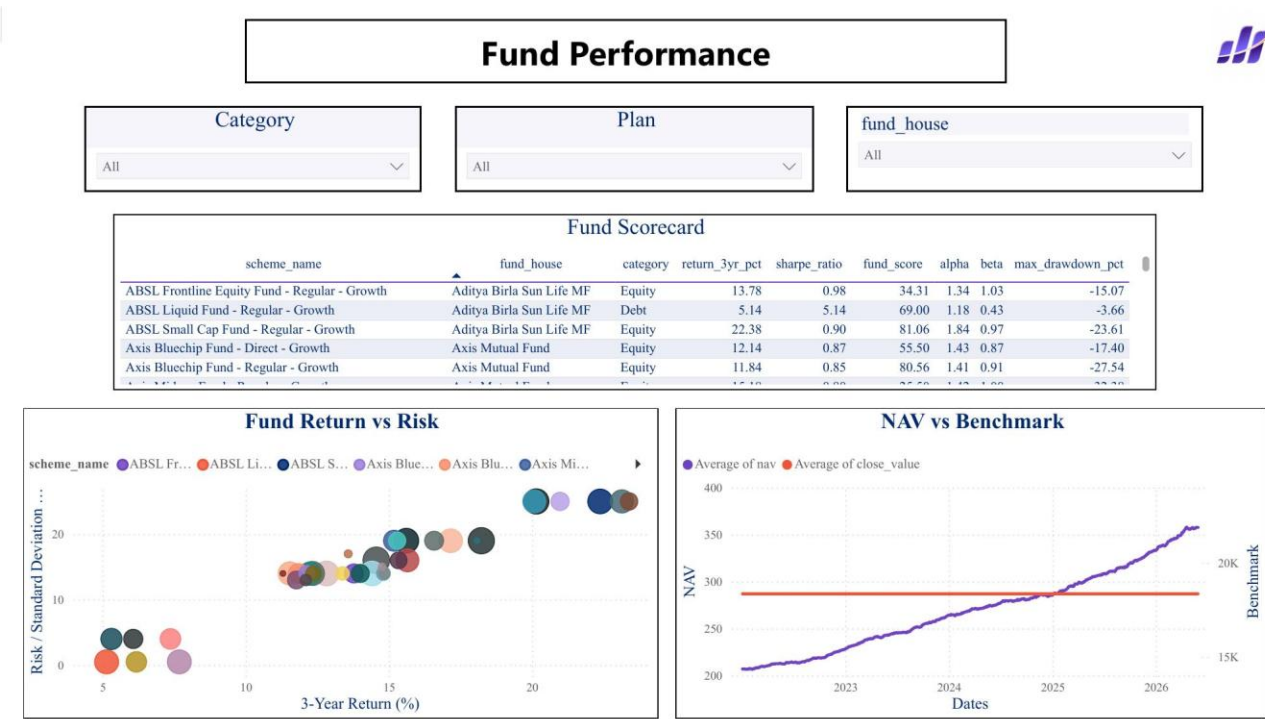
The Power BI dashboard (bluestock_mf_dashboard.pbix) delivers four analytical report pages plus a NAV Detail drill-through page, built directly on the warehouse and processed data. Filters for category, plan, fund house, age group, city tier, and state are wired across the relevant pages.

8.1 Industry Overview



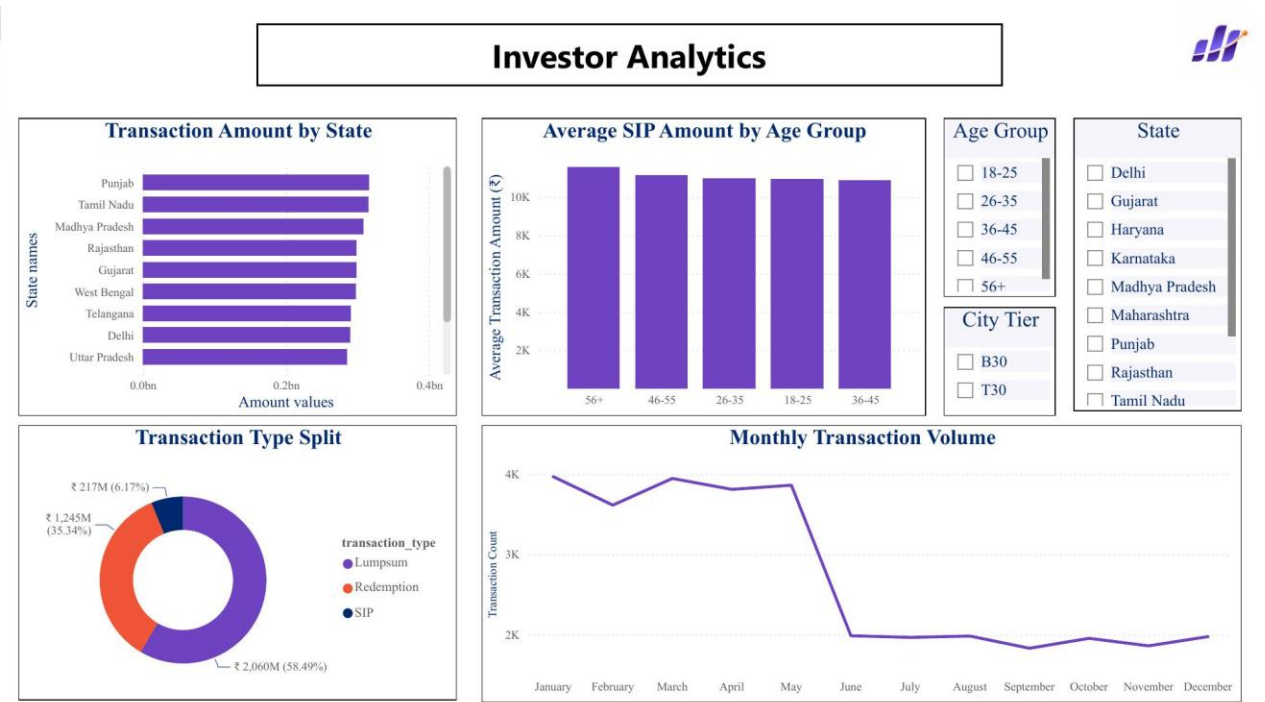
Total AUM ₹62.74 lakh crore · SIP Inflows ₹31K crore · Total Folios 26.12 crore · Total Schemes 1,522.

8.2 Fund Performance



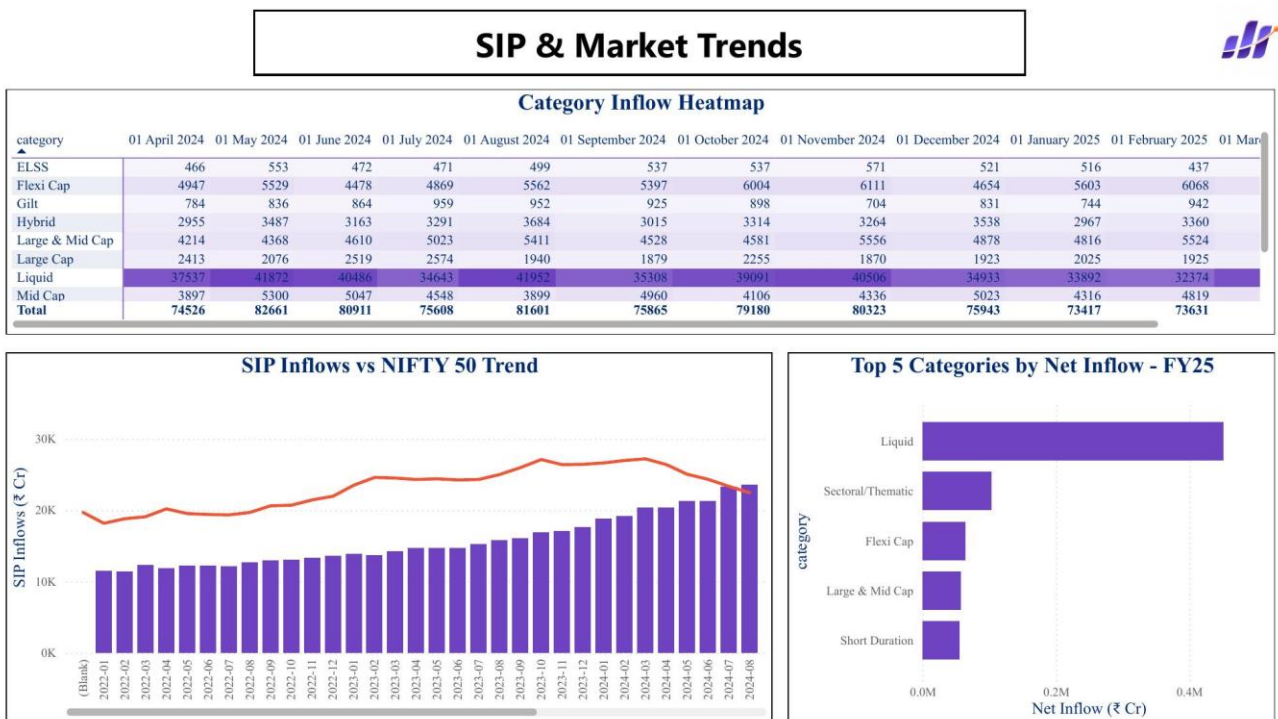
Filterable Fund Scorecard, Fund Return-vs-Risk scatter, and NAV-vs-Benchmark trend, scoped to the 40-scheme analytical sample.

8.3 Investor Analytics



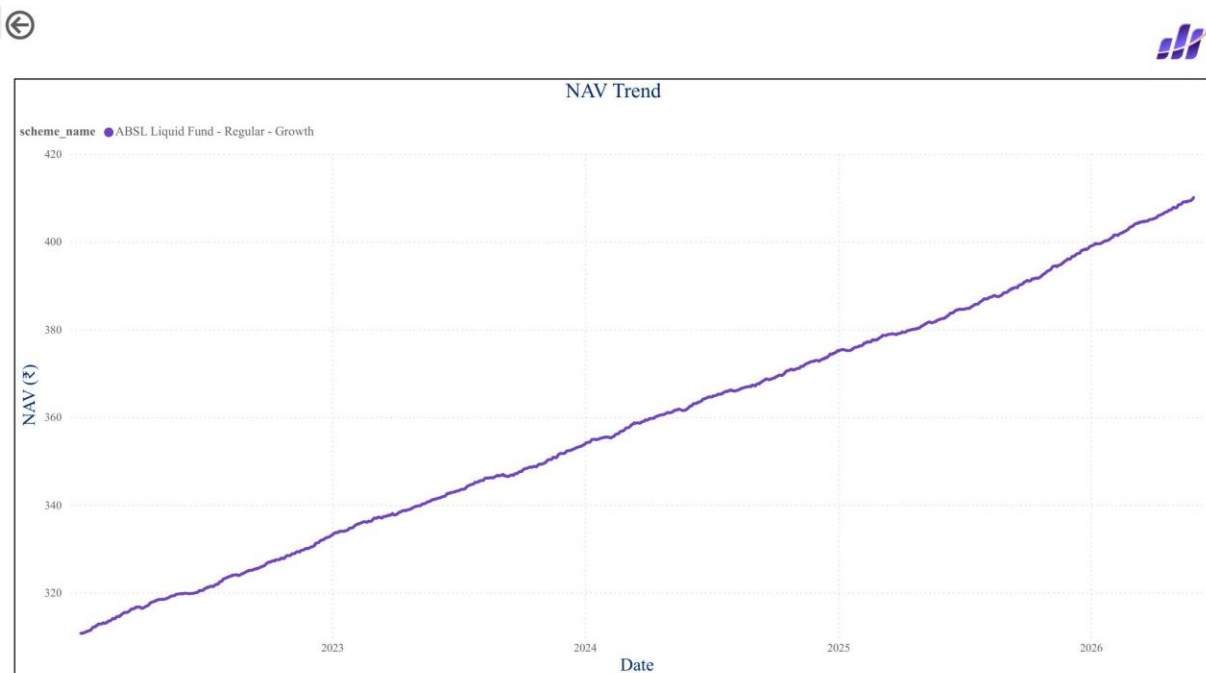
Transaction amount by state, average transaction by age group, transaction-type split, and monthly transaction volume.

8.4 SIP & Market Trends



Category inflow heatmap, SIP inflows vs. NIFTY 50 trend, and top 5 categories by FY25 net inflow.

8.5 NAV Detail (Drill-Through)



Per-scheme NAV trend, reached via drill-through from the Fund Performance scorecard (shown here for ABSL Liquid Fund – Regular – Growth).

9. Limitations

- The deep-analytics sample covers only 40 of the 1,522 schemes in the industry-wide fund master and should not be treated as a complete representation of the entire mutual-fund market.
- Historical VaR and CVaR are backward-looking and depend on the selected observation period and return distribution; both will shift as more NAV history accumulates.
- The SIP continuity flag uses a fixed 35-day average-gap rule and is not a validated churn-prediction model — only 2.2% of eligible investors clear this bar, suggesting the threshold may be tighter than typical monthly SIP cadences.
- The recommender ranks funds using risk grade and Sharpe ratio only; it does not incorporate investor suitability, goals, liquidity needs, tax considerations, or portfolio interactions.
- Sector HHI figures are carried over from the source project analysis and were not independently recomputed here, since the underlying portfolio-holdings dataset was not available in this session.
- Benchmark comparisons (alpha, beta) use NIFTY100 uniformly across all 40 schemes, including small-cap, mid-cap, and debt schemes for which a category-specific benchmark would be more appropriate.
- Investor transaction data spans roughly 17 months (Jan 2024 – May 2025) while NAV history spans over four years (Jan 2022 – May 2026); behavioural analyses that relate investor activity to fund performance are constrained to the shorter, overlapping window.
- Dashboard KPI formatting can round large counts for card-visual readability; a prior dashboard export displayed the scheme count as an approximate '2K' rather than the precise 1,522 — the current dashboard build (Section 8.1) displays the exact figure.
- The project is an analytical capstone and is not intended to provide regulated investment advice.

10. Recommendations & Conclusion

Recommendations

- Use downside-risk metrics such as VaR and CVaR alongside return and Sharpe ratio when evaluating funds, rather than return or Sharpe in isolation.
- Monitor small-cap funds closely, since the highest negative VaR and CVaR values in this dataset are concentrated in that segment.
- Use investor cohort analysis to distinguish contribution size (average SIP) from accumulated investment behaviour (total invested), which move independently across cohorts.
- Revisit the SIP continuity gap threshold (currently 35 days) against actual scheme-level SIP cadences before treating the at-risk flag as an operational signal, and build a follow-up retention model for flagged investors.
- Extend the recommender with investor goals, time horizon, drawdown tolerance, and portfolio diversification rather than Sharpe ratio alone.
- Obtain and load the portfolio-holdings dataset into the analytical warehouse so sector HHI can be recomputed and monitored on the same cadence as the other risk metrics.

- Introduce category-specific benchmarks (e.g., NIFTY Smallcap 250, NIFTY Midcap 150) for alpha/beta calculations instead of a single NIFTY100 benchmark for all schemes.
- Automate the full ETL and analytics workflow through a master pipeline script so the dashboard can be refreshed consistently rather than rebuilt manually.

Conclusion

The Bluestock Mutual Fund Analytics project demonstrates a complete business-intelligence workflow: data ingestion and validation, cleaning, exploratory analysis, fund performance measurement, advanced risk analytics, investor behaviour analysis, recommendation logic, and interactive dashboard delivery. The combination of Python analytics and Power BI provides both reproducibility for analysts and accessible reporting for business users.

Independent verification during preparation of this report confirmed that the project's headline figures, VaR/CVaR results, investor cohort statistics, SIP continuity analysis, and fund recommender outputs all reproduce exactly from the underlying data — a strong signal of analytical integrity across the pipeline. The one figure this report could not independently verify (sector HHI) is clearly flagged rather than silently repeated, consistent with the standard applied to every other number in this document.